A quadratic random-polytope upper bound

A substantial answer to Remark 4.5 of arXiv:2009.12929

Automated mathematical discovery run `fa_banach_001`, lane 9

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Candidate substantial partial result. Let $K_N = \operatorname{conv}\{\pm x_1, \dots, \pm x_N\}$, where the x_i are independent uniform points of $\mathbb{S}^{n-1}$, and let X_N have unit ball K_N. For every fixed $n \geq 3$ and $A > 0$, with probability at least $1 - N^{-A}$, every hyperplane $Y \subset X_N$ has projection constant at most

$$1 + C_{n,A} \left(\frac{\log N}{N} \right)^{2/(n-1)} .$$

This improves the source's simultaneous upper bound $1 + 2N^{-1/(2(n-1))}$. The key is a quadratic, rather than linear, conversion from spherical covering radius to Euclidean inradius.

Source question and transcription

The source studies lower bounds for norms of hyperplane projections in spaces whose unit balls are random symmetric spherical polytopes. Its Remark 4.5 also asks for good upper bounds on the hyperplane projection constants. It obtains, with probability tending to one,

$$\lambda(Y, X_N) \leq 1 + 2N^{-1/(2(n-1))}$$

simultaneously for every hyperplane Y, then states that it does not know how close the resulting range is to optimal and expects significant improvement.

Original source image

Remark 4.5. We were interested only in lower bounds on the norms of the hyperplane projections. In the context of random symmetric spherical polytopes with $2N$ vertices, establishing some good upper bounds on the projection constants of hyperplanes, also seems to be an interesting and non-trivial problem. Here we briefly explain, how to get a simple uniform upper bound that goes to 1 and that holds with probability tending to 1, as $N \to \infty$. Indeed, by a result of [6], we know that N random points in $\mathbb{S}^n$, form an $N^{-\frac{1}{2(n-1)}}$−net with the probability tending to 1, as $N \to \infty$ (see the comment before Lemma 3.2). If this condition is satisfied, then from Lemma 2.1 it follows that $\left(1 - N^{-\frac{1}{2(n-1)}}\right) \mathbb{S}_n \subseteq K$, where $K = \operatorname{conv}\{\pm x_1, \dots, \pm x_N\}$. Hence, if we denote by $X = (\mathbb{R}^n, \|\cdot\|_X)$ the normed space, for which $B_X = K$, then

$$\|x\|_2 \leq \|x\|_X \leq \frac{1}{1 - N^{-\frac{1}{2(n-1)}}}\|x\|_2 \leq \left(1 + 2N^{-\frac{1}{2(n-1)}}\right)\|x\|_2.$$

Now, let $Y \subseteq X$ be an arbitrary hyperplane. If $P : \mathbb{R}^n \to Y$ is an orthogonal projection, then $\|P\|_2 = 1$ and from the estimates above it easily follows that $\|P\|_X \leq 1 + 2N^{-\frac{1}{2(n-1)}}$. Thus, combining this with Theorem 1.2 we get, that for $N \to \infty$, the projection constant of an arbitrary hyperplane in X lies in the interval

$$[1 + c_n N^{-(2n^2+4n+6)}, 1 + 2N^{-\frac{1}{2(n-1)}}],$$

with the probability tending to 1. We do not know, to what extent this range is best possible, but it is likely that it could be improved (restricted) significantly.

Source: T. Kobos, arXiv:2009.12929v3, PDF p. 12, Remark 4.5. This is a direct crop from `source_paper.pdf`.

Definitions

For a hyperplane Y in a normed space X, its relative projection constant is

$$\lambda(Y, X) = \inf\{\|P\|_{X\to X} : P^2 = P,\ \operatorname{ran} P = Y\}.$$

For a finite set $F \subset \mathbb{S}^{n-1}$, its Euclidean covering radius is

$$R(F) = \sup_{u\in\mathbb{S}^{n-1}} \min_{x\in F} \|u - x\|_2 .$$

Surface area measure on $\mathbb{S}^{n-1}$, normalized to have mass one, is denoted by σ.

Idea / proof intuition

The source converts an ε-net into the rough inclusion $(1-\varepsilon)B_2^n \subset K_N$. But all vertices lie on the sphere. If $\|u - x\|_2 \leq \varepsilon$, then exactly

$$\langle u, x\rangle = 1 - \frac{1}{2}\|u - x\|_2^2 \geq 1 - \frac{\varepsilon^2}{2}.$$

Thus the correct inradius loss is quadratic: $(1 - \varepsilon^2/2)B_2^n \subset K_N$.

The random covering radius is at most a constant times $(\log N/N)^{1/(n-1)}$ with arbitrarily high polynomial probability. A short cap-measure and union-bound argument proves precisely what is needed. Squaring this radius gives the displayed projection-constant rate.

Deterministic quadratic conversion

Lemma 1 (Covering radius to simultaneous projection bounds)**.** *Let $F \subset \mathbb{S}^{n-1}$, let $K = \mathrm{conv}(F \cup -F)$, and assume $R(F) \le \varepsilon < \sqrt{2}$. If X is the normed space with unit ball K, then every hyperplane $Y \subset \mathbb{R}^n$ satisfies*

$$\lambda(Y, X) \le \frac{1}{1 - \varepsilon^2/2}. \tag{1}$$

In particular, if $\varepsilon \le 1$, then $\lambda(Y, X) \le 1 + \varepsilon^2$.

Proof. For every $u \in \mathbb{S}^{n-1}$, choose $x \in F$ with $\|u - x\|_2 \le \varepsilon$. The identity above gives

$$h_K(u) = \max_{z \in F} |\langle u, z\rangle| \ge \langle u, x\rangle \ge 1 - \varepsilon^2/2.$$

Domination of support functions is equivalent to containment of convex bodies, so

$$(1 - \varepsilon^2/2) B_2^n \subset K \subset B_2^n. \tag{2}$$

Write $r = 1 - \varepsilon^2/2$. The corresponding norms obey

$$\|v\|_2 \le \|v\|_X \le r^{-1} \|v\|_2 .$$

For a hyperplane Y, let Q_Y be Euclidean orthogonal projection onto Y. Then

$$\|Q_Y v\|_X \le r^{-1} \|Q_Y v\|_2 \le r^{-1} \|v\|_2 \le r^{-1} \|v\|_X .$$

Since Q_Y is an admissible projection, this proves (1). Finally, $(1 - s)^{-1} \le 1 + 2s$ for $0 \le s \le 1/2$, with $s = \varepsilon^2/2$. □

A self-contained random covering estimate

Put $d = n - 1$. There is a constant $\kappa_n > 0$ such that every Euclidean spherical cap

$$C(z, t) = \{u \in \mathbb{S}^{n-1} : \|u - z\|_2 \le t\}, \qquad 0 < t \le 1,$$

satisfies

$$\sigma(C(z, t)) \ge \kappa_n t^d. \tag{3}$$

For example one may take

$$\kappa_n = \frac{2^{n-2}}{(n-1)\pi^{n-1}}.$$

Indeed, the angular radius is $2 \arcsin(t/2) \ge t$, and integrating $\sin^{n-2}\theta \ge (2\theta/\pi)^{n-2}$ for $\theta \le \pi/2$ proves (3).

Lemma 2 (Polynomial-tail covering bound)**.** *Let $x_1, \dots, x_N$ be independent and uniform on $\mathbb{S}^{n-1}$, and set $F_N = \{x_1, \dots, x_N\}$. For $0 < t \le 1$,*

$$\mathbb{P}\{R(F_N) > t\} \le \kappa_n^{-1} \left(\frac{4}{t}\right)^d \exp\left(-\kappa_n N (t/2)^d\right). \tag{4}$$

Proof. Take a maximal $t/2$-separated set $Z \subset \mathbb{S}^{n-1}$. It is a $t/2$-net, while the caps $C(z, t/4)$, $z \in Z$, are disjoint. By (3),

$$|Z| \le \kappa_n^{-1} (4/t)^d. \tag{5}$$

If every cap $C(z, t/2)$ contains one sample point, then F_N is a t-net by the triangle inequality. Each such cap is empty with probability at most

$$(1 - \kappa_n (t/2)^d)^N \le \exp(-\kappa_n N (t/2)^d).$$

The union bound and (5) give (4). □

Main result

Theorem 1 (Improved random upper bound)**.** *Fix $n \geq 3$ and $A > 0$. There are $C_{n,A}, N_{n,A} > 0$ such that, for every $N \geq N_{n,A}$, the random space*

$$B_{X_N} = K_N = \mathrm{conv}\{\pm x_1, \dots, \pm x_N\}$$

satisfies, with probability at least $1 - N^{-A}$,

$$\sup_{\substack{Y \subset X_N \\ \dim Y = n-1}} \lambda(Y, X_N) \leq 1 + C_{n,A} \left(\frac{\log N}{N} \right)^{2/(n-1)}. \tag{6}$$

Proof. In (4), take

$$t_N = \left(\frac{2^d (A+2)}{\kappa_n} \frac{\log N}{N} \right)^{1/d}.$$

For all sufficiently large N, $t_N \leq 1$, and substitution into (4) gives

$$\mathbb{P}\{R(F_N) > t_N\} \leq \frac{2^d}{(A+2)\log N} N^{-(A+1)} \leq N^{-A}.$$

On the complementary event, the deterministic lemma applies simultaneously to every hyperplane and yields

$$\lambda(Y, X_N) \leq 1 + t_N^2.$$

This is (6), with

$$C_{n,A} = \left(\frac{2^{n-1}(A+2)}{\kappa_n} \right)^{2/(n-1)}.$$

□

Comparison, verification, and scope

The source upper excess is $2N^{-1/(2(n-1))}$. The new excess is

$$C_{n,A} (\log N/N)^{2/(n-1)}.$$

For fixed n, their ratio is $O(N^{-3/(2(n-1))} (\log N)^{2/(n-1)})$. The theorem also quantifies probability tending to one by an arbitrary polynomial tail.

- The spherical identity, support-function containment, and norm comparison in (2) are exact.
- The random estimate uses only a deterministic net, a cap-volume lower bound, independence, and a union bound.
- The same random points, on the same event, work for all hyperplanes; there is no union bound over the uncountable Grassmannian.
- Symmetrization causes no loss: a net made by the x_i alone is already sufficient for $K_N = \mathrm{conv}\{\pm x_i\}$.

Upgrade attempt and obstruction. The logarithm can be removed from estimates at a fixed direction, but a simultaneous inradius argument must control the largest empty spherical cap; random covering radii naturally live at the $(\log N/N)^{1/(n-1)}$ scale. A log-free simultaneous projection bound would therefore need information about projection constants not mediated by global Euclidean inradius. No matching lower bound for the random projection excess was obtained; the source's lower bound remains much smaller.

Novelty audit. Cheap run indexes and bounded primary-source/arXiv searches through 17 August 2026 found the source paper and general random-sphere covering work, including arXiv:1512.07470, but no later statement combining the quadratic support-function conversion with simultaneous hyperplane projection constants. Novelty confidence is moderate, not definitive.

Limitations. This substantially improves the upper endpoint of the source's interval but does not determine its optimal order and does not improve the source's lower endpoint. Constants are explicit but deliberately crude. Dimension is fixed while $N \to \infty$, as in the source remark.

Human review. Pending. Priority checks are the cap-measure normalization in (3), the support-function implication in (2), and whether an equivalent projection bound has appeared outside the arXiv searches.

References

[1] T. Kobos, *A uniform lower bound on the norms of hyperplane projections of spherical polytopes*, arXiv:2009.12929v3, especially Lemma 2.1 and Remark 4.5.

[2] J. S. Brauchart, A. B. Reznikov, E. B. Saff, I. H. Sloan, Y. G. Wang, and R. S. Womersley, *Random point sets on the sphere—hole radii, covering, and separation*, arXiv:1512.07470.